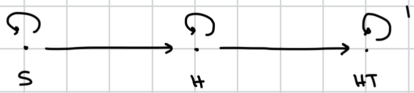
ON COMPUTING THE NUMBER OF FLIPS

[HENCE ON THE NUMBER OF JUMPS IN A MARKOV CHAIN]

- EXPECTED NUMBER OF FLIPS / JUMPS

  - SEQUENCE HT



[ALL OTHER PROB = $\frac{1}{2}$]

LET E_x BE THE EXPECTED NUMBER OF FLIPS FROM x TO HT

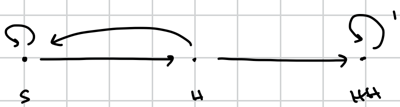
$$(\cdot)\quad \begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \\ E_{HT} = 0 \end{cases}$$

SOLVING THE SYSTEM WE HAVE $E_S = 4$

$\downarrow$

INCIDENTALLY THIS IS ALSO THE INVERSE OF THE PROB $\frac{1}{4}$ OF FLIPPING THE SEQUENCE HT AMONG THE SEQUENCES OF LENGTH 2

  - SEQUENCE HH



[ALL OTHER PROB = $\frac{1}{2}$]

LET E_x BE THE EXPECTED NUMBER OF FLIPS FROM x TO HH

$$(\cdot)\quad \begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HH} \\ E_{HH} = 0 \end{cases}$$

SOLVING THE SYSTEM WE HAVE $E_S = 6$

$\downarrow$

THIS IS NOT THE INVERSE OF THE PROB $\frac{1}{4}$ OF FLIPPING THE SEQUENCE HT AMONG THE SEQUENCES OF LENGTH 2

=) WITH THE SYSTEM OF EQUATIONS [E_x] WE MEASURE THE NUMBER OF FLIPS BEFORE WE SEE A SEQUENCE FOR THE FIRST TIME

=) FIRST PASSAGE TIME

IN GENERAL: E_x = EXPECTED NUMBER OF FLIPS FROM x to *SEQUENCE*
= FIRST PASSAGE TIME FROM x to *SEQUENCE*

$$E_x = 1 + \sum_j P_{xj} E_j \qquad \forall x \qquad [E_{*SEQUENCE*} = 0]$$

=) WITH THE INVERSE OF THE PROBABILITY WE MEASURE THE NUMBER OF FLIPS BEFORE WE SEE THE SAME SEQUENCE AGAIN

=) RECURRENCE TIME

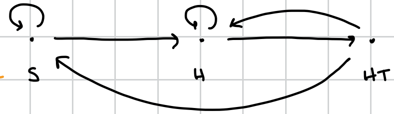
- RECURRENCE TIME : IT'S A DIFFERENT MARKOV CHAIN [WITH A LINK BACK TO START].
WE DO NOT NEED A SYSTEM OF EQUATIONS BUT "JUST A FORMULA"

R_x = RECURRENCE TIME FOR STATE x

$$R_x = 1 + \sum_j P_{xj} E_j \qquad [E_j = \text{FIRST TIME PASSAGE FROM } j \text{ to } x]$$

- SEQUENCE HT

MC 2



[ALL PROB = $\frac{1}{2}$]

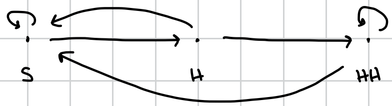
$E_S = 4 \qquad E_H = 2 \qquad E_{HT} = 0 \qquad$ [SOLVING (•)]

$$R_{HT} = 1 + \sum_j P_{HT,j} E_j \qquad [P_{HT,j} \text{ ACCORDING to MC 2}]$$

$$= 1 + P_{HT,S} E_S + P_{HT,H} E_H + P_{HT,HT} E_{HT}$$

$$= 1 + \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 2 + 0 \cdot 0 = 1 + 2 + 1 = 4 \qquad [QED]$$

- SEQUENCE HH

MC2


[ALL PROB $= \frac{1}{2}$]

$E_S = 6 \quad E_H = 4 \quad E_{HH} = 0$ [SOLVING (•)]

$$R_{HH} = 1 + \sum_j P_{HH,j} E_j$$ [$P_{HH,j}$ ACCORDING to MC2]

$$= 1 + P_{HH,S} E_S + P_{HH,H} E_H + P_{HH,HH} E_{HH}$$

$$= 1 + \frac{1}{2} \cdot 6 + 0 \cdot E_H + \frac{1}{2} \cdot 0 = 1 + 3 = 4$$ [QED]

INTUITION: HH AND HT HAVE THE SAME RECURRENCE TIME [BOTH SEQUENCES of LENGHT 2] BUT TO SEE HH FOR THE FIRST TIME WE NEED MORE FLIPS. WHY? BECAUSE IN HH, H RE-OCCUR INSIDE THE SEQUENCE IT SELF.

$S \rightarrow H \longrightarrow HH$